



BIT N BUILD '26 · GUJARAT ROUND PRESENTATION

EcoTrack AI

AI-Powered Waste Management & Recycling
Optimizer

Problem Statement: Smart City Urban Waste Management, Real-Time IoT
Telemetry & Dynamic Route Optimization

TEAM

TriByte



Jemit Vaghasiya

Team Leader & Backend Architect



Jay Sohaliya

Frontend & UI/UX Specialist



Yug Bhensadaiya

AI/ML & Database Engineer



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Why waste operations need a rethink

The Problem

Operations rely on static schedules and manual logs, with no integrated real-time platform.



Frequent Bin Overflows

Bins overflow before scheduled pickups, causing health hazards, odors and pollution.



High Fuel & Operational Costs

Fixed routes visit empty bins while missing critical high-fill locations.



Inefficient Waste Sorting

No real-time classification delays recycling and raises cross-contamination.



No Multi-Role Coordination

Managers, drivers and analysts lack one unified real-time dashboard.

Target Users



Waste Managers / Admins

Municipal supervisors directing citywide operations, fleet dispatch and bin deployment.



Collection Drivers

Field workers following turn-by-turn navigation and completing stop manifests.



Operations Analysts

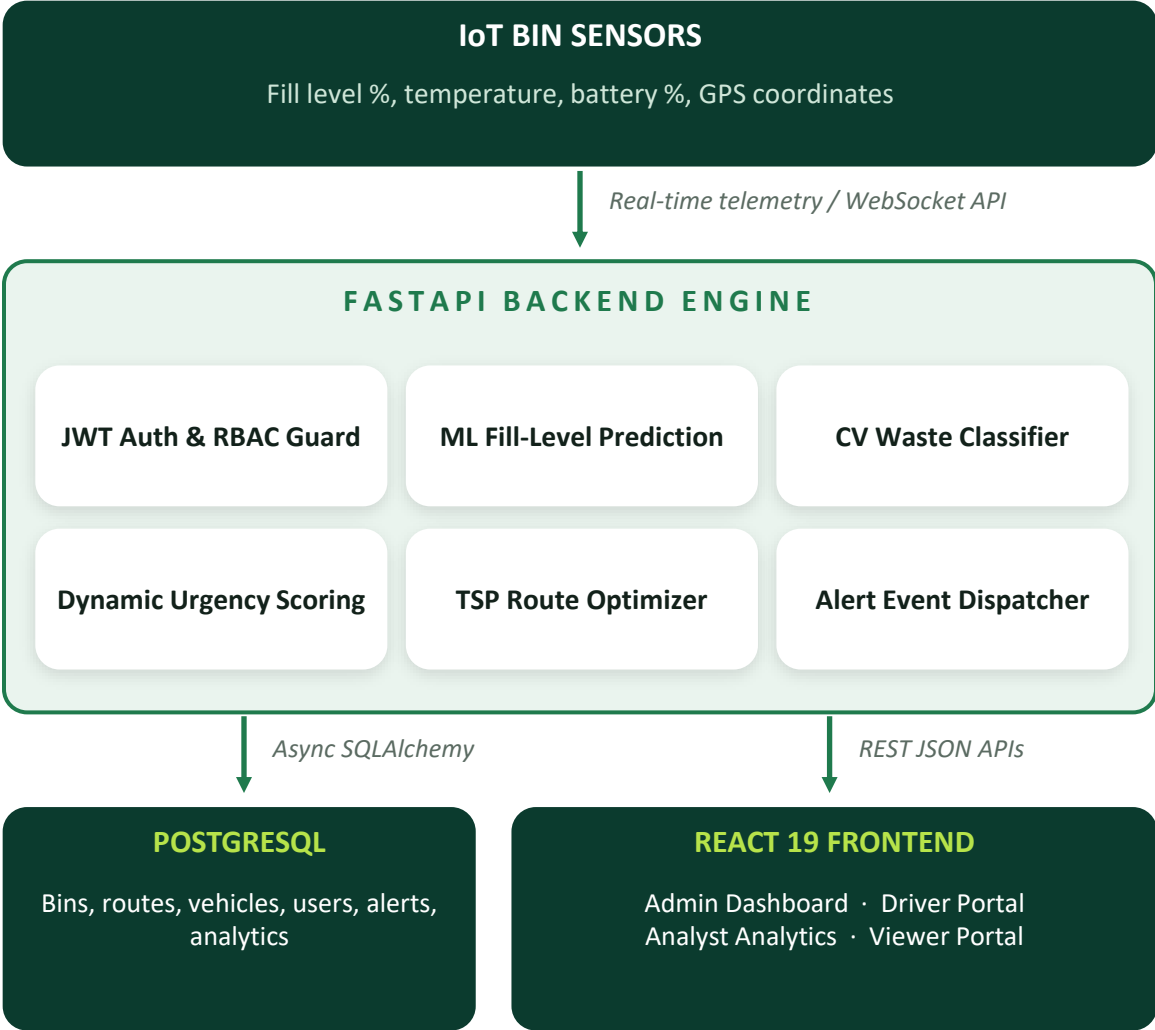
Urban planners monitoring waste trends, recycling purity and fleet efficiency.

PROPOSED SOLUTION

EcoTrack AI ingests real-time IoT telemetry, predicts fill levels with ML, classifies waste with computer vision, scores urgency dynamically and optimizes driver pickup manifests.

Built on a modern async stack

Languages	Python 3.10+, TypeScript, JavaScript, HTML5, CSS3
Frontend	React 19, Vite, Tailwind CSS, Lucide React, Leaflet / OpenStreetMap GIS
Backend	FastAPI (async REST engine), Pydantic v2
Database	PostgreSQL, SQLAlchemy (async ORM), Alembic migrations
AI / ML	Linear & polynomial regression, computer vision classification pipeline
Security	JWT, Passlib (Bcrypt), RBAC security scoping
Dev & Testing	Pytest, Postman, Git/GitHub, Docker, Uvicorn



A data-driven workflow, end to end

1

Telemetry Ingestion

Ingests live fill-level, temperature, battery % and GPS data from IoT sensors.

2

Predictive Analytics

ML forecasting calculates fill rates (%/hr) and estimates hours until overflow.

3

Dynamic Priority Scoring

Urgency scores from fill level (>85% critical), risk factor (temperature/fire) and zone density.

4

Route Optimization

A Traveling Salesperson (TSP) heuristic builds distance-optimized manifests for each vehicle.

5

Driver Dispatch & Execution

Turn-by-turn portal guides each stop, runs a live timer, resets the bin to 0% on collection and flags skipped stops in yellow.



Integration & Automated Testing

Automated Pytest Suite

End-to-end backend tests for Auth, Bins, Routes, Vehicles, Alerts and User Management (pytest tests/).

Frontend Compilation

Clean Vite + TypeScript build passing with exit code 0.

What we have built



Smart Bin Management

Live GIS map of bin capacity, battery, temperature and overflow-risk badges.



ML Prediction & Alerts

Fill-level forecasting with alerts for high temperature, sensor dropout and overflow risk.



Collection Planning

Multi-route optimization and driver dispatch manifest generation.



Driver Portal & Live Mode

Stop-by-stop navigation, live timer, green active stop, 0% reset, yellow skip.



AI Waste Classification

Sorts waste into Organic, Recyclable and Hazardous with confidence scores.



Fleet & Vehicle Tracking

Truck tracking, fuel level, capacity utilization % and maintenance logs.



Analytics & Export

Interactive analytics charts and customizable CSV report export.



Multi-Tier RBAC

Distinct secure experiences for Admin, Driver, Analyst and Viewer roles.

IMPLEMENTED OUTCOMES

100%

Fully functional, end-to-end stack ready for deployment

~35%

Projected fuel reduction by skipping low-fill bins and optimizing routes

Zero

Preventable overflows: alerts fire before the 85% threshold

Who built what

JV

Jemit Vaghasiya

Team Leader & Backend Architect

FastAPI architecture, database schemas, JWT RBAC security, route optimization, database seeding and API routing.

JS

Jay Sohaliya

Frontend & UI/UX Specialist

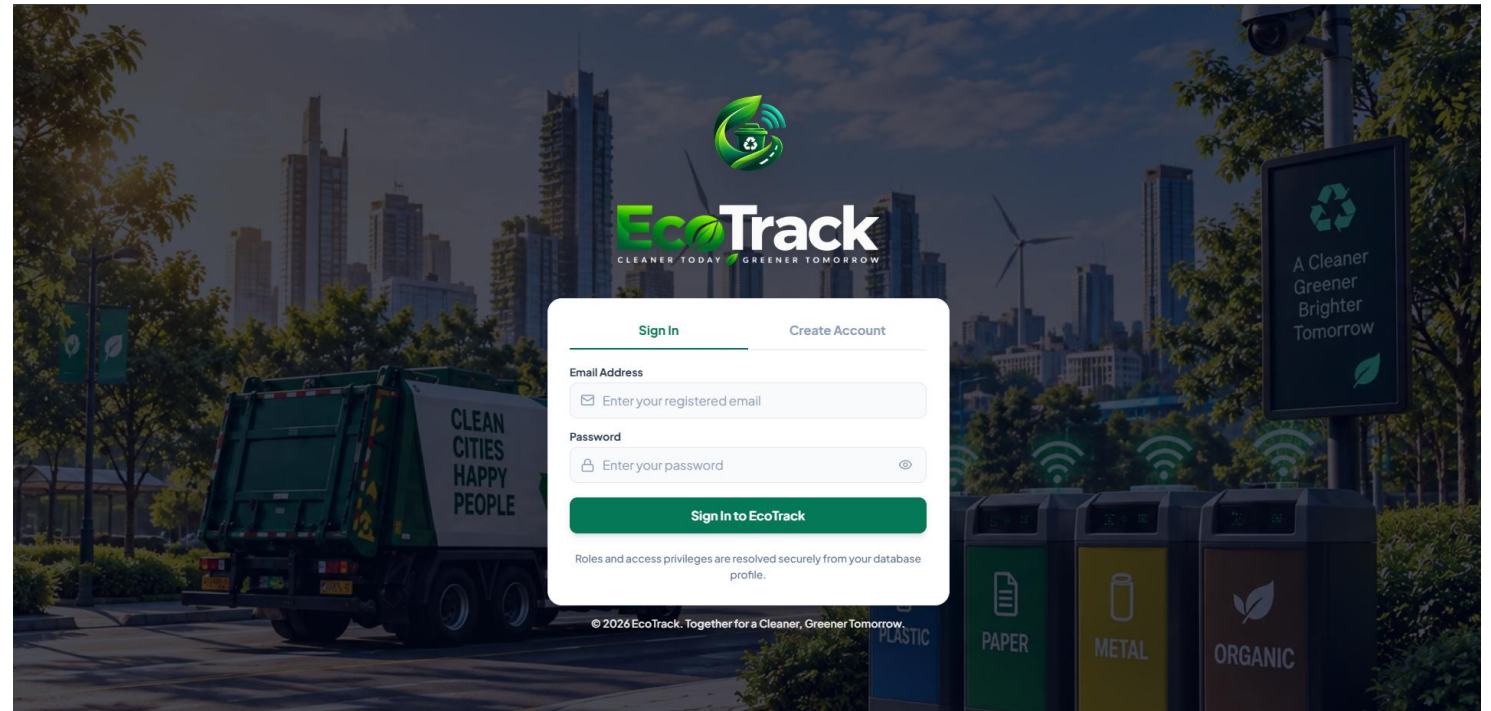
React 19 + Vite UI, Leaflet GIS map, Driver Portal live collection workflow and responsive styling.

YB

Yug Bhensadaiya

AI/ML & Database Engineer

ML fill-level prediction, computer vision pipeline, PostgreSQL migrations and the Pytest suite.



From smarter bins to smarter cities

Real-World Impact



Environmental Improvement

Reduces urban bin overflows, mitigates fire hazards and eliminates unpleasant odors.



Economic Savings

Cuts fuel and maintenance expenditure by ~35% through targeted collection routes.



Public Health & Hygiene

Keeps city sectors clean with automated, proactive alert dispatches.

Future Scope



Hardware IoT Prototyping

Custom LoRaWAN / NB-IoT sensor hardware deployment.



On-Truck Edge AI

Truck-mounted computer vision for automated contamination checks.



Citizen App

Citizens report overflowing bins and earn recycling reward points.

PROJECT LINKS

Explore the project



GitHub Repository

<https://github.com/Vaghasiya-Jemit-kanaiyalal/Bit-n-Build-26-TriByte>



Project Documentation

README.md (included in the repository root)



Demo Video

Available in the <https://youtu.be/rfD1JiGouYI?si=1o2CltjoljiQDWWs>

Thank You

Team TriByte

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